

# Final Assignment: Credit Portfolio Risk

## 1 Assignment Description

### Objective

Analyze the **one-year credit risk** of given portfolios using simulation.

You will:

- clean real-world-style messy data
- construct a modeling dataset
- estimate model inputs
- simulate portfolio losses
- interpret and compare results

### Portfolios

You must analyze:

- **Portfolio 1**
- **Portfolio 2**

**Optional (advanced):**

- **Portfolio 3**

### Data

You are given multiple raw datasets:

- `entities_info.csv` — firm characteristics
- `portfolio_*.csv` — portfolio holdings
- `stock_returns.csv` — firm returns
- `industry_index_returns.csv` — industry returns
- `default_history_20y.csv` — historical defaults

The data are intentionally **messy and incomplete**.

### Part 1 — Data Cleaning

Prepare a clean dataset suitable for modeling.

Minimum expectations:

- standardize **entity codes** and **industry labels**
- handle **missing values** (e.g., LGD, EAD, returns, defaults)
- remove or resolve **duplicates**
- ensure **consistent date formats**
- merge data into a usable modeling table

Clearly explain your cleaning choices.

## Part 2 — Model Setup and Parameter Estimation

Choose **one or two models** discussed in the course.

For your chosen model(s):

- identify required inputs (e.g., PD, LGD, dependence structure)
- estimate parameters using the data
- justify assumptions or approximations

There is no single correct approach, clarity and reasoning are key.

## Part 3 — Simulation and Risk Measurement

Using your model(s):

- simulate the **one-year loss distribution**
- compute risk measures such as:
  - Expected Loss (EL)
  - standard deviation
  - Value-at-Risk (e.g., 99%)
  - Expected Shortfall

You are free to choose the simulation design and level of complexity.

## Part 4 — Analysis and Interpretation

Discuss your results. Possible directions:

- Compare **Portfolio 1 vs Portfolio 2**
- Compare results across models (if applicable)
- Discuss the result
- (Optional) Analyze **Portfolio 3**

Focus on **clear reasoning and economic intuition**.

## Deliverables

Submit:

- **Python code**
- **Short report (3 - 5 pages)** structured in an academic format, including:
  - **Data:** description of the dataset, cleaning procedures, and assumptions
  - **Methodology:** model framework, parameter estimation, and relevant literature
  - **Results and Discussion:** analysis and interpretation of results

# 2 Grading Rubric

## Overall Grade Distribution

- Code: 30%

- Report: 60%
- Oral Exam: 10%

### Important Note on the Oral Exam

The oral exam serves as a **verification of authorship and understanding**. Students are expected to explain their code, modeling choices, and results.

- If a student is unable to demonstrate sufficient understanding of their submission, the assignment will be considered **failed**.
- The oral exam is not intended as a standalone assessment, but as a tool to ensure that the work is **original and not fully AI-generated**.

### General Requirements (applies to all components)

- Work must be **original**. Students must be able to explain and justify their code and modeling choices.
- **Fully AI-generated submissions will be considered a failure.**

## 1. Code

### *Essential requirements*

- Code runs without errors
- Clear structure with logical segmentation
- Sufficient comments explaining key steps
- Student demonstrates understanding of the implementation

### *Excellent*

- Clean and efficient code structure
- Use of reusable functions and modular design
- Clear and well-organized documentation

## 2. Model and Analysis

### *Essential requirements*

- At least one model is implemented correctly
- Clear assumptions are stated and justified
- Basic explanation of the model type (statistical understanding)
- Appropriate risk metrics are computed (e.g., EL, VaR, ES)

### *Excellent*

- Two models are implemented, justified, and compared
- Strong understanding of the underlying statistical concepts
- Multiple risk metrics are used and compared
- Clear interpretation of differences across models and portfolios

## 3. Report (3–5 pages, excluding references)

### *Essential requirements*

- Written in an academic style (no AI-generated text)
- Structured into:
  - Data
  - Methodology
  - Results and Discussion
- Length between 3 and 5 pages (excluding references)
- Proper use of references and citation style

*Excellent*

- Clear and coherent academic writing
- Sections are well integrated (not treated as isolated parts)
- Demonstrates strong understanding of the statistical concepts
- Results are clearly interpreted and linked to the methodology